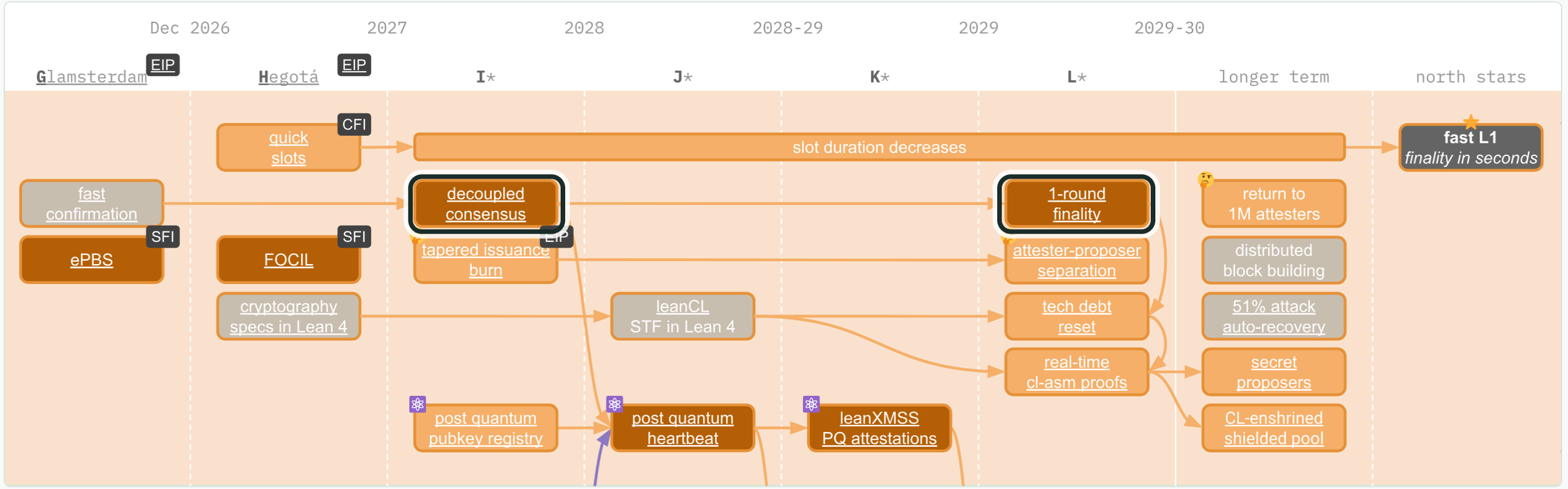
One-round finality

Future of fast finality

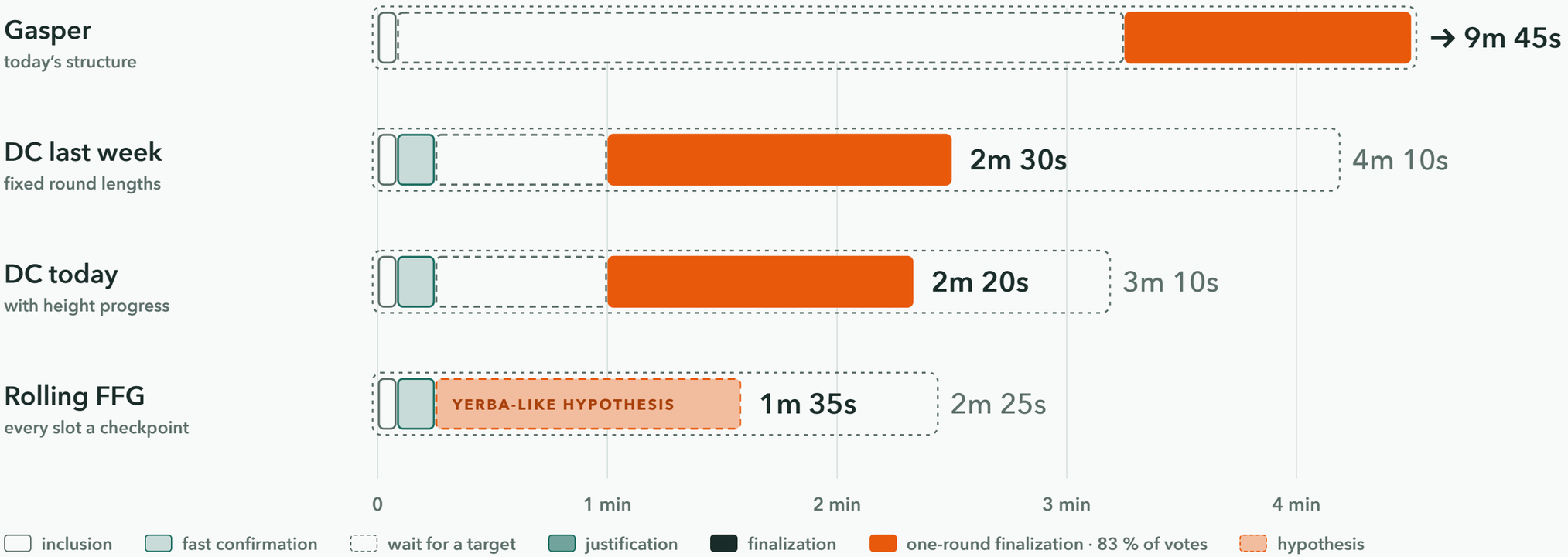
FAST FINALITY WORKSHOP · CAMBRIDGE · THURSDAY 15 OCTOBER · 14:00

Strawmap



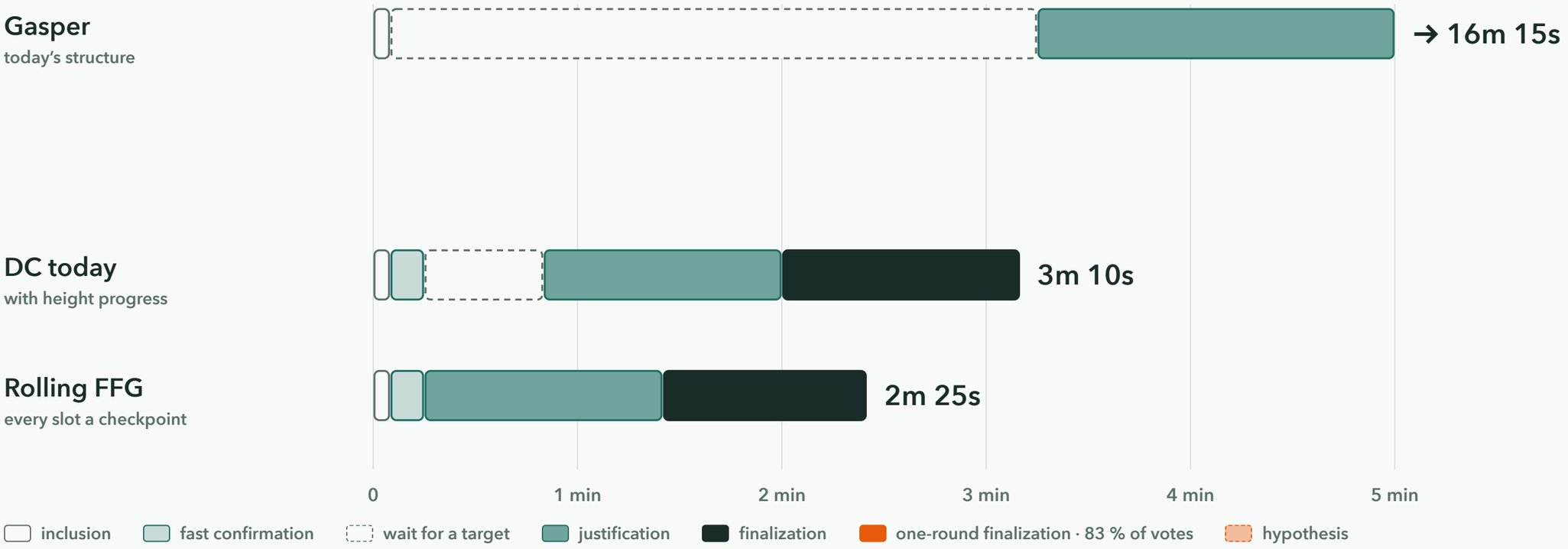
strawmap.org · consensus layer · updated 9 Oct 2026

Time to finality



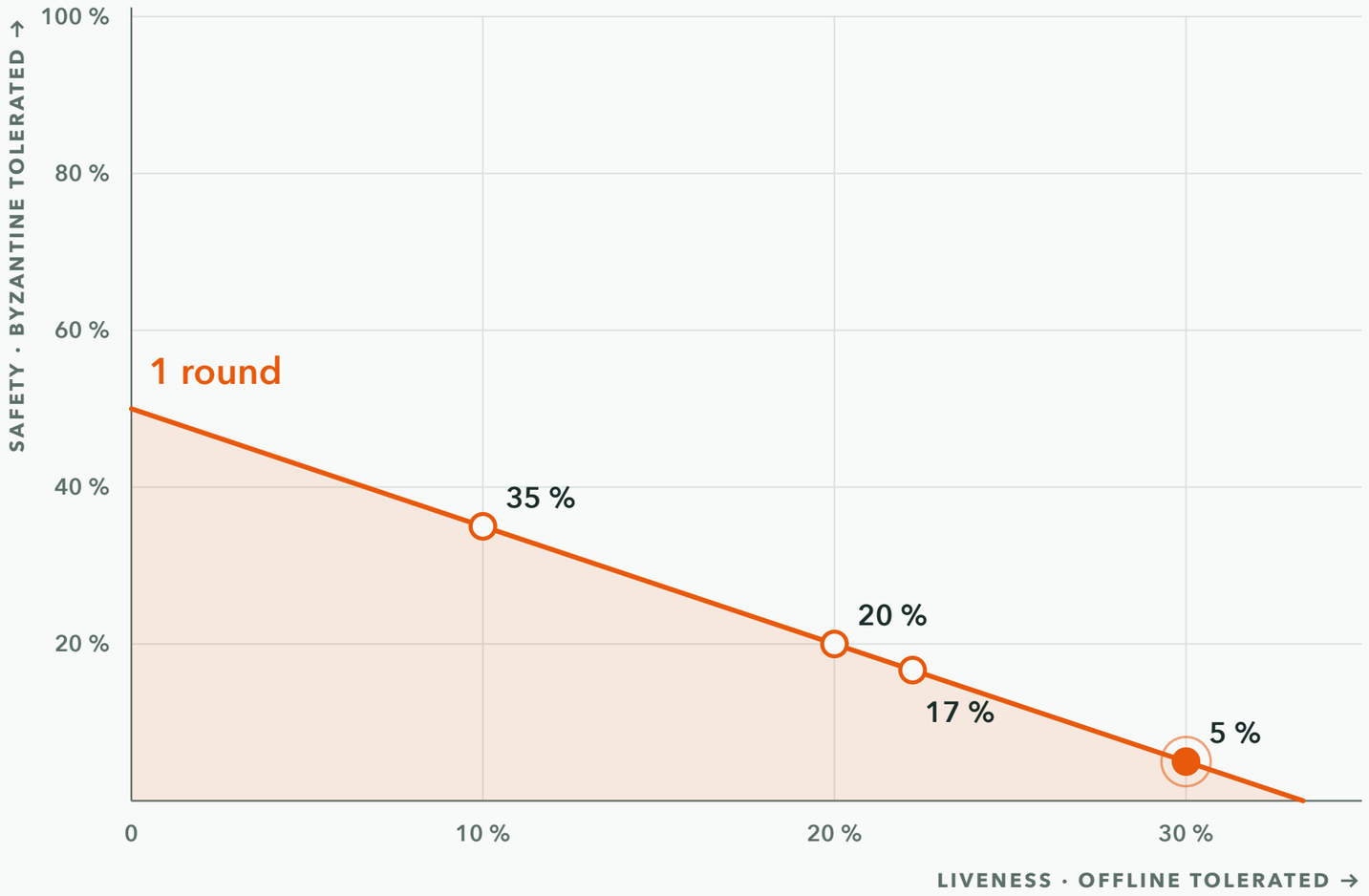
Expected values · finality.watch model · 1.1M validators · 10-second slots

Turn the knobs



Expected values · finality.watch model · full-set round: DC 10 slots (1m 40s), Gasper 39 slots (6m 30s)

One round: safety vs. liveness



EVERYONE

finalize at 70 %

safe 5 % · live 30 %

2t + 3p < n: Lamport ’02, Dutta–Guerraoui–Vukolić ’05

NEW PAPER

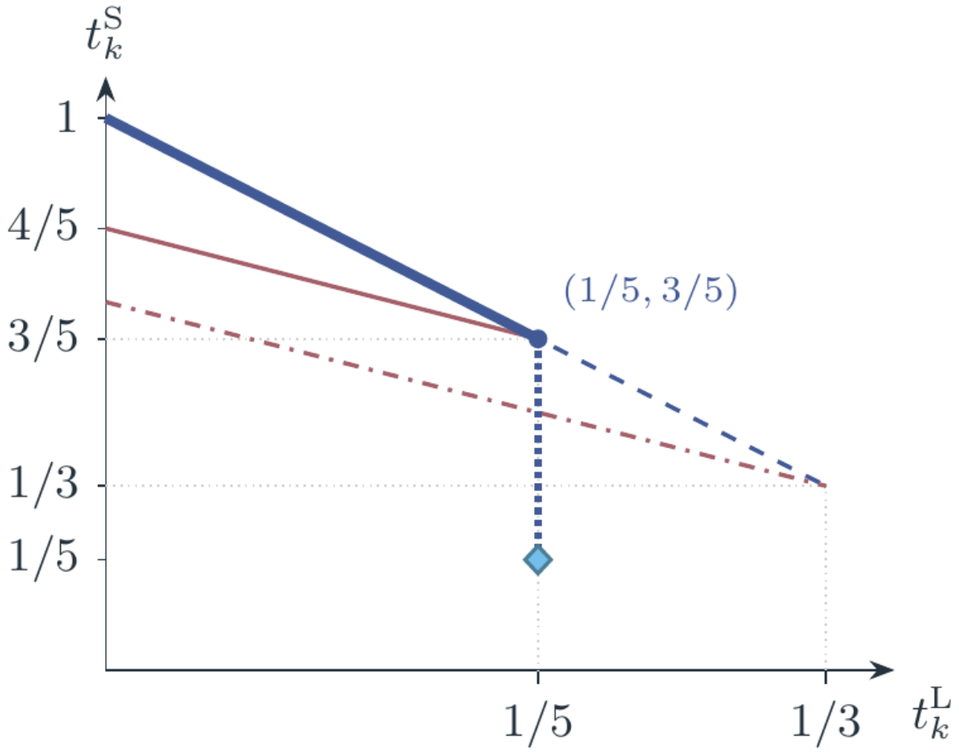
Safety gadget

Short Paper: Flexible Consensus in Two and Three Rounds

Abstract. We apply OFlex's checkpointing-style safety gadget on top of an off-the-shelf base consensus protocol like Minimmit with two-round good-case latency. This yields two interesting results. First, from a practical perspective, two-round protocols suffer from lower 20% resilience (compared to 33% for regular three-round protocols), which the gadget makes up for: it gives clients the option to trade off an extra round of latency for safety resilience up to 100%, or to remain at two rounds and 20% resilience. Second, from a theoretical perspective, the resulting flexible consensus protocol supports a substantial segment of the optimal safety–liveness tradeoff in three rounds, improving over the state-of-the-art four rounds of the generic OFlex construction over a three-round base protocol. This is a first step towards the round complexity of optimal flexible consensus for safety–liveness resilience tradeoff, which remains an interesting open problem.

Keywords: Flexible consensus · Byzantine fault tolerance · Round complexity

Client trade-offs

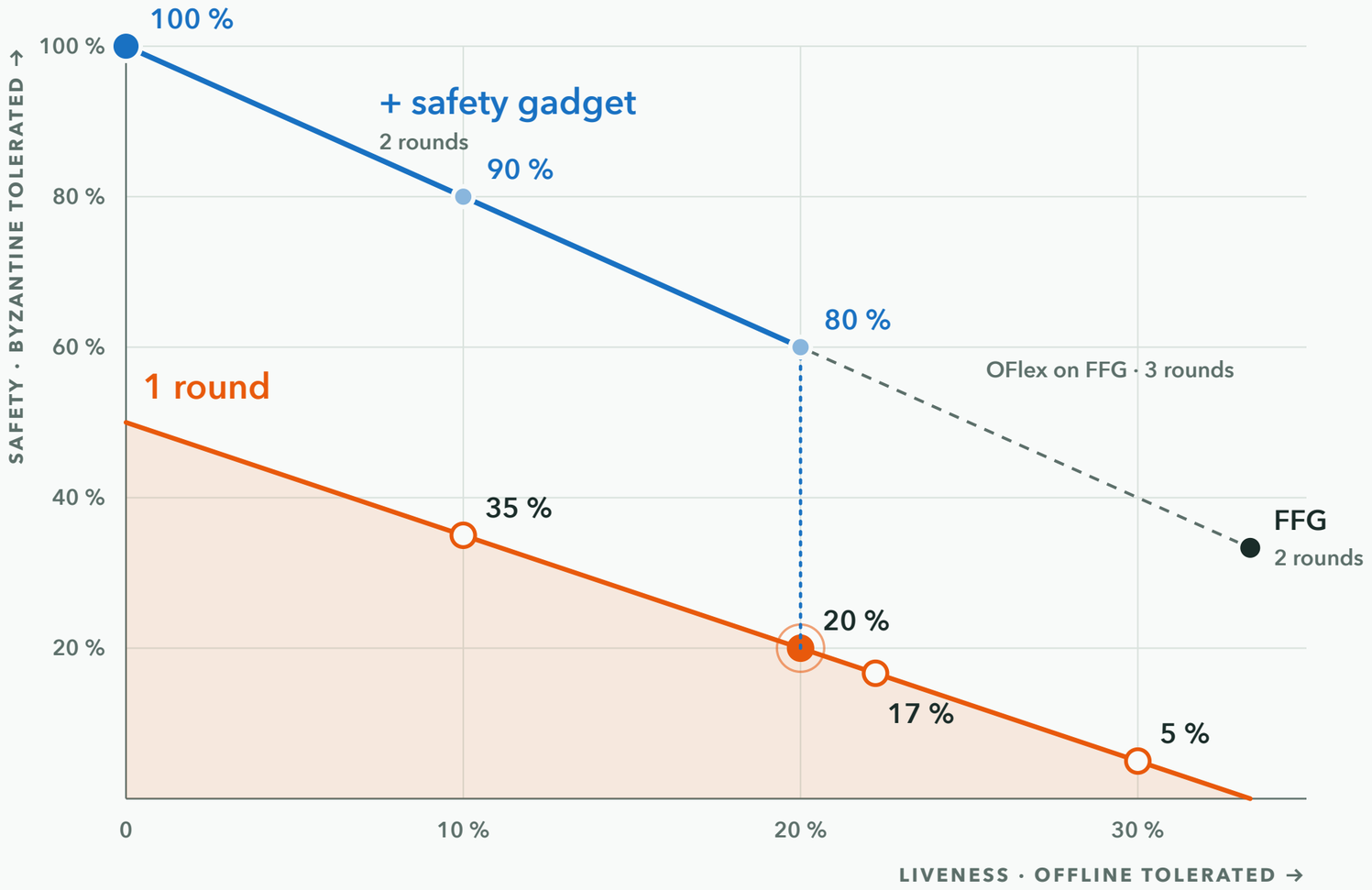


Good-case latency

- Base protocol + gadget: 3δ
- Lower safety targets: 3δ
- OFLEX (PBFT): 4δ
- FBFT, $q_r = 4/5$: 3δ
- FBFT, $q_r = 2/3$: 3δ
- Base client: 2δ

The solid diagonal lies on OFLEX's boundary. Each curve fixes one server protocol.

Safety gadget



EVERYONE

finalize at 80 %

safe 20 % · live 20 %

OBSERVERS · YOU CHOOSE

wait for 80 %

safe 60 % · live 20 %

wait for 90 %

safe 80 % · live 10 %

wait for 100 %

safe 100 % · live 0 %

gadget frontier: OFlex ‘24

Three signals

